

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
7				Indicative content • crude oil is a complex mixture of hydrocarbon compounds with a range of boiling points • the useful products from fractional distillation are called fractions • crude oil is boiled / vaporised / evaporated before it enters the fractionating column • fractions condense at different heights inside the column because they have different boiling points • the higher the boiling point, the lower down it condenses • boiling points are linked to the length of the carbon chain – the longer the chain, the higher the boiling point • compounds in the same fraction have similar boiling points	6			6		

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					AO1	AO2	AO3	Total	Maths	Prac
				5-6 marks Detailed explanation including reference to changes of state and the link between chain length and boiling point of fractions <i>There is a sustained line of reasoning which is coherent, relevant, substantiated and logically structured. The candidate uses appropriate scientific terminology and accurate spelling, punctuation and grammar.</i> 3-4 marks Basic explanation with some reference to changes of state, chain length and boiling point of fractions <i>There is a line of reasoning which is partially coherent, largely relevant, supported by some evidence and with some structure. The candidate uses mainly appropriate scientific terminology and some accurate spelling, punctuation and grammar.</i> 1-2 marks Some knowledge of changes of state, chain length or boiling point of fractions <i>There is a basic line of reasoning which is not coherent, largely irrelevant, supported by limited evidence and with very little structure. The candidate uses limited scientific terminology and inaccuracies in spelling, punctuation and grammar.</i> 0 marks <i>No attempt made or no response worthy of credit.</i>						
				Question 7 total	6	0	0	6	0	0